

FX Equity Spillover (Working Draft)

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1 Abstract

The goal of this model is to analyze and develop a strategy for trading FX using equity returns. There is a relationship between foreign equity returns and their currency. While the relationship between a country's equity returns and their corresponding currencies has been discussed, this work does contribute or focus to the economic interpretation or consequences. Instead, it investigates whether the relationship can be effectively modeled and exploited for trading.¹

2 Historical Data

The initial model was developed using FX futures collected from Bloomberg Terminal and constructed using front month roll adjustments. Although FX futures do not fully capture the FX market, they provide sufficient cross-sectional coverage for these models. FX Futures were selected because they provide a convenient instrument for both historical backtesting and real-world trading execution. From a backtesting perspective, FX futures are advantageous since they embed the carry accrued from holding the underlying foreign currency, simplifying data handling, especially for non-deliverable forward currencies.² From an implementation standpoint, FX futures are easily financeable, of highly liquid pairs, and have high-quality data.

Some of the FX futures have low liquidity which is taken into consideration. All currency futures are offered by CME, which tends to provide relatively reasonable liquidity. Bloomberg total return values are collected as well, and Bloomberg offers more FX returns series than futures. The cross-correlation between the matched FX Futures and Bloomberg Carry tickers are quite high. Asian currencies tend to have lower cross-correlations between FX Future and Bloomberg Carry, which could be from index construction, but they are also the series with the least amount of data amongst the Bloomberg Carry indices. With regards to the Bloomberg FX Carry indices liquidity can't be determined and since there isn't a centralized market the distinction will be separated by G10 rates and Emerging Currencies. On the following page is table of the FX Futures and Bloomberg Carry dates.

¹A GitHub repo with the full codebase and results can be found <https://github.com/diegoalvarez/FXEquitySpillover>

²See strategy ten in *Advanced Futures Trading Systems* by Rob Carver for a clear explanation of carry in FX futures.

Future	Name	Liquidity	Start Date	End Date
AD1 Curncy	Australian Dollar Future	Liquid	1987-01-13	2026-01-12
BP1 Curncy	British Pound Future	Liquid	1986-05-28	2026-01-12
BR1 Curncy	Brazilian Real Future	Liquid	1995-11-09	2026-01-12
CD1 Curncy	Canadian Dollar Future	Liquid	1986-04-04	2026-01-12
EC1 Curncy	Euro Foreign Future	Liquid	1998-05-20	2026-01-12
JY1 Curncy	Japanese Yen Future	Liquid	1986-05-23	2026-01-12
KO1 Curncy	Korean Won Future	Illiquid	2006-09-19	2026-01-12
NO1 Curncy	Norwegian Krone Future	Illiquid	2002-05-17	2026-01-12
NV1 Curncy	New Zealand Dollar Future	Liquid	1997-05-08	2026-01-12
PE1 Curncy	Mexican Peso Future	Liquid	1995-04-26	2026-01-12
PP1 Curncy	Polish Zloty Future	Illiquid	2004-07-12	2026-01-12
SE1 Curncy	Swedish Krona Future	Illiquid	2002-05-17	2026-01-12
SF1 Curncy	Swiss Franc Future	Liquid	1986-04-07	2026-01-12
SIR1 Curncy	Indian Ruppee Future	Illiquid	2013-01-29	2026-01-12
UCA1 Curncy	Offshore Chinese Renminbi Future	Illiquid	2012-09-18	2026-01-12

Table 1: FX Futures Historical Data

Country	Index Name	Group	Start Date	End Date
Argentina	Argentine Peso Carry Return Index	EM Carry	1999-01-04	2026-03-31
Australia	Australian Dollar Carry Return Index	G10 Carry	1990-01-02	2026-03-31
Brazil	Brazilian Real Carry Return Index	EM Carry	1999-01-04	2026-03-31
UK	British Pound Carry Return Index	G10 Carry	1990-01-02	2026-03-31
Canada	Canadian Dollar Carry Return Index	G10 Carry	1990-01-02	2026-03-31
Chile	Chilean Peso Carry Return Index	EM Carry	1999-01-04	2026-03-31
China	Chinese Renminbi Carry Return Index	EM Carry	1998-12-14	2026-03-31
Colombia	Colombian Peso Carry Return Index	EM Carry	1999-01-04	2026-03-31
Denmark	Danish Krone Carry Return Index	EM Carry	1990-01-02	2026-03-31
Euro	Euro Carry Return Index	G10 Carry	1990-01-02	2026-03-31
Hong Kong	Hong Kong Dollar Carry Return Index	EM Carry	1990-01-02	2026-03-31
Iceland	Iceland Krona Carry Return Index	EM Carry	1999-01-04	2026-03-31
India	Indian Rupee Carry Return Index	EM Carry	1998-12-31	2026-03-30
Indonesia	Indonesian Rupiah Carry Return Index	EM Carry	1999-01-04	2026-03-31
Japan	Japanese Yen Carry Return Index	G10 Carry	1990-01-02	2026-03-31
Malaysia	Malaysian Ringgit Carry Return Index	EM Carry	1999-01-04	2026-03-31
Mexico	Mexican Peso Carry Return Index	EM Carry	1999-01-04	2026-03-31
New Zealand	New Zealand Dollar Carry Return Index	G10 Carry	1990-01-02	2026-03-31
Norway	Norwegian Krone Carry Return Index	G10 Carry	1990-01-02	2026-03-31
Peru	Peruvian Sol Carry Return Index	EM Carry	1999-01-04	2026-03-31
Philippines	Philippine Peso Carry Return Index	EM Carry	1999-01-04	2026-03-31
Poland	Polish Zloty Carry Return Index	EM Carry	1999-01-04	2026-03-31
Russia	Russian Ruble Carry Return Index	EM Carry	1999-01-04	2026-03-31
Singapore	Singapore Dollar Carry Return Index	EM Carry	1999-01-04	2026-03-31
South Africa	South African Rand Carry Return Index	EM Carry	1990-01-02	2026-03-31
Korea	South Korean Won Carry Return Index	EM Carry	1999-01-04	2026-03-31
Sweden	Swedish Krona Carry Return Index	EM Carry	1990-01-02	2026-03-31
Switzerland	Swiss Franc Carry Return Index	EM Carry	1990-01-02	2026-03-31
Taiwan	Taiwanese Dollar Carry Return Index	EM Carry	1999-01-04	2026-03-31
Thailand	Thai Baht Carry Return Index	EM Carry	1999-01-04	2026-03-31
Turkey	Turkish Lira Carry Return Index	EM Carry	1999-01-04	2026-03-31

Table 2: Bloomberg FX Carry Total Return Indices Historical Data

To model the equity component of the futures ETFs and Equity benchmarks are used. The natural selection is to use the VanEck ETFs and their respective benchmarks. Their historical data is collected as well as other attributes of the indices. Below is a table of the mapping between countries and their respective ETFs and Benchmarks.

Country	ETF Ticker	Benchmark
Argentina	ARGT US Equity	M1AAR5R Index
Australia	EWA US Equity	NDDUAS Index
UK	EWU US Equity	NDDUUK Index
Brazil	EWZ US Equity	M1BR2550 Index
Canada	EWK US Equity	M1CXBLK Index
Chile	ECH US Equity	M1CL5IM Index
China	MCHI US Equity	M7CN Index
Colombia	COLO US Equity	M1CXGXB Index
Denmark	EDEN US Equity	M1DK5IM Index
Euro	EZU US Equity	NDDUEMU Index
Hong Kong	EWK US Equity	NU730026 Index
Iceland	GLCR US Equity	MVISLGTR Index
Israel	EIS US Equity	MISCNU Index
India	INDA US Equity	NDEUSIA Index
Indonesia	EIDO US Equity	NU724020 Index
Japan	EWJ US Equity	NDDUJN Index
Korea	EWY US Equity	M1KR2550 Index
Norway	ENOR US Equity	M1NO5IM Index
Malaysia	EWM US Equity	NDDUMAF Index
New Zealand	ENZL US Equity	NU756513 Index
Mexico	EWK US Equity	M1MX5IM Index
Poland	EPOL US Equity	M1PL5IM Index
Peru	EPU US Equity	M1PECAPD Index
Philippines	EPHE US Equity	NU710465 Index
Sweden	EWK US Equity	M1CXBISA Index
Russia	ERUS US Equity	MXRU500I Index
Singapore	EWS US Equity	M1CXBISC Index
South Africa	EZA US Equity	M1CXBAC Index
Switzerland	EWL US Equity	M1CH2550 Index
Taiwan	EWT US Equity	M1CXBISD Index
Thailand	THD US Equity	M1TH5IM Index
Turkey	TUR US Equity	NU724019 Index

Table 3: Country ETF and Benchmark Index Matching

Type	Fields
Equity	Cur. Mkt. Cap, Day to Day Ret. Gross Dividend, Fund Flow, Volume
Index	FCF (Est.), CUR EV/EBITDA (Est.), Div. Yield (Est.), EBIT (Est.), BEST_EPS, Gross Margin (Est.), P/E (Est.), P/Sales (Est.), Target Price (Est.), Cur. EV, EBIT, EBITDA, FCF Yield, Gros Margin, EPS, Net. Agg Div Yld, P/E Ratio, P/Sales

Table 4: Fundamental Fields by Type

The cross correlation between the matched Bloomberg FX Carry Indices and FX Futures are highly correlated. The Bloomberg FX Carry is done in long dollar format as opposed to the FX Futures. This leads to an artificial negative correlation. The returns are also volatility-targeted at a 10% annualized volatility target with lagged and contemporaneously hedge returns. Below is the table of correlation amongst the matched FX futures and Bloomberg Carry.

Country	Lagged	Contemporaneous	None
Australia	-0.90	-0.93	-0.91
Brazil	-0.68	-0.84	-0.84
Canada	-0.91	-0.92	-0.91
China	0.69	0.65	0.64
Euro	-0.95	-0.95	-0.95
India	-0.68	-0.67	-0.65
Japan	-0.95	-0.95	-0.94
Korea	-0.65	-0.66	-0.70
Mexico	-0.90	-0.91	-0.89
New Zealand	-0.89	-0.89	-0.89
Norway	-0.93	-0.93	-0.93
Poland	-0.95	-0.95	-0.95
Sweden	-0.93	-0.94	-0.91
Switzerland	-0.95	-0.94	-0.94
UK	-0.96	-0.96	-0.96

Table 5: Correlation between FX Futures and Bloomberg Carry Index

The correlations are quite close which is a good sign, and much of the lower correlation pairs come from a shorter history. The table below compares the number of days between FX Futures and Carry indices.

Country	Carry	Future
Australia	9456	9841
Brazil	7107	7597
Canada	9456	10044
China	7122	3334
Euro	9456	7007
India	7108	3266
Japan	9456	9995
Korea	7107	4825
Mexico	7107	7753
New Zealand	9456	7248
Norway	9456	5955
Poland	7107	5417
Sweden	9456	5959
Switzerland	9456	10026
UK	9456	9999

Table 6: Number of Days in Dataset

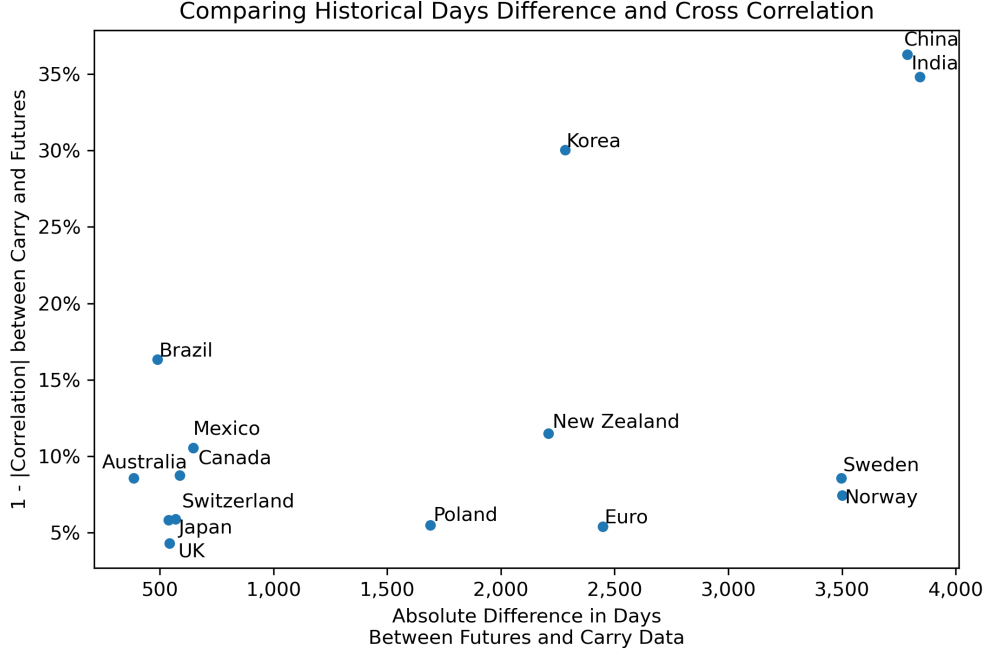


Figure 1: Correlation vs. Days Spread

3 Theoretical Approach

Following Cenedese et. al [2] the theoretical foundation around this model follows a no-arbitrage constraint. For a domestic and foreign equity return R_d and R_f where the foreign return has to be scaled by some currency s_t

$$\mathbb{E}(R_d \cdot m_h) = \mathbb{E}\left(R_f \cdot \frac{s_{t+1}}{s_t} \cdot m_h\right) = 1 \quad (1)$$

m_h is the discounting factor to maintain no-arbitrage also called the stochastic discount factor. Then rearrange the formula using covariance rules.

$$\mathbb{E}(R_d) \cdot \mathbb{E}(m_h) + \text{cov}(m_h, R_d) = \mathbb{E}\left(R_f \cdot \frac{s_{t+1}}{s_t}\right) \cdot \mathbb{E}(m_h) + \text{cov}\left(m_h, R_f \cdot \frac{s_{t+1}}{s_t}\right) = 1 \quad (2)$$

Then let $\mathbb{E}(m_h)$ be the one-period, risk free zero-coupon bond R_b in country d .

$$\mathbb{E}(R_d) \cdot \frac{1}{R_b} + \text{cov}(m_h, R_d) = \mathbb{E}\left(R_f \cdot \frac{s_{t+1}}{s_t}\right) \cdot \frac{1}{R_b} + \text{cov}\left(m_h, R_f \cdot \frac{s_{t+1}}{s_t}\right) = 1 \quad (3)$$

The equation above is useful since it separates the contribution of risk premia and the risk free rate. In a no-arbitrage setting, the stochastic discount factor m_h discounts the returns back to one. If the discounting is too slow - corresponding to positive excess returns - the covariance term $\text{cov}(m_h, R_d)$ adjusts to offset these returns back to one. Conversely, if discounting is too fast, implying negative excess returns, the covariance term adjusts in the opposite direction. Then re-arrange the formula to separate the equity returns (excess returns) and the risk premia

$$\underbrace{\mathbb{E}\left(R_f \cdot \frac{s_{t+1}}{s_t}\right) \frac{1}{R_b} - \mathbb{E}(R_d) \cdot \frac{1}{R_b}}_{\text{excess equity return differential}} = \underbrace{\text{cov}(m_h, R_d) - \text{cov}\left(m_h, R_f \cdot \frac{s_{t+1}}{s_t}\right)}_{\text{difference in risk premia}} \quad (4)$$

Now, to fully isolate the currency returns, use a similar approach for log returns. Therefore,

$$\mathbb{E}\left(\exp\left(\log\left(m_h R_f \frac{s_{t+1}}{s_t}\right)\right)\right) = \mathbb{E}(\exp(\log(m_h R_d))) \quad (5)$$

Equation 5 can be rewritten as

$$\mathbb{E}(\exp(\tilde{m} + r_f + \Delta s_{t+1})) = \mathbb{E}(\exp(\tilde{m} + r_d)) \quad (6)$$

Where r_d , r_f , and Δs_{t+1} are the log domestic and foreign equity with changes in the currency, $\tilde{m}$ is $\log(m_h)$. Then with some variance and covariance rules we can get the log-comparable to equation 3.

$$\mathbb{E}(\tilde{m}) + \mathbb{E}(r_f) + \mathbb{E}(\Delta s_{t+1}) + \frac{1}{2}\text{var}(\tilde{m} + r_f + \Delta s_t) = \mathbb{E}(\tilde{m}) + \mathbb{E}(r_d) + \frac{1}{2}\text{var}(\tilde{m} + r_d) \quad (7)$$

Then re-arrange

$$\mathbb{E}(r_f) + \mathbb{E}(\Delta s_{t+1}) - \mathbb{E}(r_d) = \frac{1}{2}(\text{var}(\tilde{m} + r_f + \Delta s_t) - \text{var}(\tilde{m} + r_d)) \quad (8)$$

$$= \frac{1}{2}(\text{var}(r_f + \Delta s_t) - \text{var}(r_d)) + \text{cov}(\tilde{m}, r_f + \Delta s_{t+1}) - \text{cov}(\tilde{m}, r_d) \quad (9)$$

Which results in an equation very similar to equation 3 but in log form

$$\mathbb{E}(\Delta s_{t+1}) = \underbrace{\mathbb{E}(r_d) - \mathbb{E}(r_f)}_{\text{expected log return differential}} + \underbrace{\text{cov}(\tilde{m}, r_f + \Delta s_{t+1}) - \text{cov}(\tilde{m}, r_d)}_{\text{difference in risk premia}} + \underbrace{\frac{1}{2}(\text{var}(r_f + \Delta s_t) - \text{var}(r_d))}_{\text{log-variance adjustment}} \quad (10)$$

The approach is appealing because it preserves flexibility in the sign of the covariance term $\text{cov}(m_h, R_j \frac{S_{t+1}}{S_t})$. Hau and Rey [4] argue that portfolio rebalancing by international investors implies a negative relationship, whereas Bohn and Tesar[1], and Froot, Scharfstein, and Stein[3] document evidence consistent with a positive relationship.

Cenedese et al further this framework beyond equity returns and use other equity factors such as dividend yield, value, and momentum. Those other factors will be considered but require empirical analysis beforehand.

4 Examining the Factor Exposure

The overall goal is to analyze the FX exposure of the returns differential between FX and equity markets, but before doing that the sovereign equity exposure must be accounted for to set a benchmark. FX returns and Equity ETF returns are volatility-targeted as well to control for heteroskedasticity. The volatility targets are set at 10% annualized volatility using an exponentially weighted standard deviation using 100 prior days. Accounting for lagged and contemporaneously hedged results is taken into consideration.

The factor exposure is measured through the β of the OLS regression. The regressions are ran instantaneously as opposed to lagged to account for the factor exposure. The returns type (No Hedging, Lagged Hedging and Perfectly Hedged) are matched between FX returns and ETF returns. On the next page is the results of the regression with the corresponding p -values and t -stat. All of them are statistically significant, and most are positive implying that increases in equities are followed by increases in FX returns.

Now that the baseline has been established a separate regression is run using the returns differential between the sovereign ETFs and SPY. On the following page is the regression results for using the returns differential between sovereign equities and SPY, controlling for all the volatility targets as well.

ETF	Country	Vol. Target FX	Coef			<i>p</i> -value			<i>t</i> -value			<i>R</i> ²		
			Lagged	Perfect	None	Lagged	Perfect	None	Lagged	Perfect	None	Lagged	Perfect	None
ARGT	Argentina	ARS Carry	0.10	0.11	0.08	0.00	0.00	0.00	6.53	6.58	4.36	0.01	0.01	0.00
COLO	Colombia	COP Carry	0.61	0.61	0.32	0.00	0.00	0.00	49.01	50.18	52.22	0.36	0.37	0.39
ECH	Chile	CLP Carry	0.61	0.62	0.27	0.00	0.00	0.00	50.96	51.27	49.67	0.36	0.36	0.35
EDEN	Denmark	DKK Carry	0.34	0.35	0.15	0.00	0.00	0.00	21.65	22.20	22.44	0.12	0.12	0.12
EIDO	Indonesia	IDR Carry	0.31	0.32	0.07	0.00	0.00	0.00	20.39	21.13	21.61	0.09	0.10	0.10
EIS	Israel	ISR Future	0.51	0.51	0.20	0.00	0.00	0.00	39.11	40.29	39.94	0.26	0.27	0.26
ENOR	Norway	NOK Carry	0.68	0.68	0.34	0.00	0.00	0.00	53.90	53.62	57.00	0.45	0.45	0.48
		NOK Future	0.66	0.66	0.33	0.00	0.00	0.00	52.64	52.43	53.92	0.44	0.44	0.45
ENZL	New Zealand	NZD Carry	0.68	0.69	0.36	0.00	0.00	0.00	58.31	58.56	56.04	0.46	0.47	0.45
		NZD Future	0.66	0.67	0.35	0.00	0.00	0.00	55.39	55.91	51.65	0.44	0.45	0.41
EPHE	Philippines	PHP Carry	0.25	0.25	0.05	0.00	0.00	0.00	16.00	16.07	14.63	0.06	0.06	0.05
EPOL	Poland	PLN Carry	0.66	0.66	0.29	0.00	0.00	0.00	54.35	54.56	61.27	0.43	0.43	0.49
		PLN Future	0.64	0.65	0.29	0.00	0.00	0.00	53.03	53.22	59.36	0.42	0.42	0.47
EPU	Peru	PEN Carry	0.41	0.40	0.09	0.00	0.00	0.00	28.49	28.48	28.40	0.16	0.16	0.16
EWA	Australia	AUD Carry	0.63	0.63	0.30	0.00	0.00	0.00	68.78	69.02	77.49	0.39	0.39	0.44
		AUD Future	0.59	0.60	0.28	0.00	0.00	0.00	62.88	63.65	65.55	0.35	0.35	0.36
EWC	Canada	CAD Carry	0.58	0.59	0.23	0.00	0.00	0.00	61.96	61.99	68.24	0.34	0.34	0.38
		CAD Future	0.54	0.55	0.21	0.00	0.00	0.00	56.18	56.81	59.58	0.30	0.30	0.32
EWD	Sweden	SEK Carry	0.47	0.48	0.20	0.00	0.00	0.00	46.21	46.57	52.38	0.22	0.22	0.27
		SEK Future	0.51	0.51	0.21	0.00	0.00	0.00	45.18	45.37	47.55	0.26	0.26	0.28
EWH	Hong Kong	HKD Carry	0.10	0.10	0.00	0.00	0.00	0.00	8.60	8.89	5.84	0.01	0.01	0.00
EWJ	Japan	JPY Carry	0.02	0.03	0.03	0.05	0.01	0.00	1.92	2.48	5.20	0.00	0.00	0.00
		JPY Future	0.03	0.04	0.04	0.00	0.00	0.00	2.96	3.35	7.70	0.00	0.00	0.01
EWL	Switzerland	CHF Carry	0.27	0.28	0.13	0.00	0.00	0.00	24.65	25.64	22.69	0.07	0.08	0.06
		CHF Future	0.27	0.28	0.13	0.00	0.00	0.00	24.56	25.40	23.03	0.07	0.08	0.07
EWM	Malaysia	MYR Carry	0.28	0.28	0.06	0.00	0.00	0.00	14.03	13.74	20.93	0.03	0.03	0.06
EWS	Singapore	SGD Carry	0.45	0.46	0.09	0.00	0.00	0.00	41.44	41.70	38.76	0.20	0.20	0.18
EWT	Taiwan	TWD Carry	0.28	0.28	0.03	0.00	0.00	0.00	22.47	23.06	19.73	0.07	0.08	0.06
EWU	UK	GBP Carry	0.40	0.40	0.16	0.00	0.00	0.00	37.31	37.15	39.15	0.16	0.15	0.17
		GBP Future	0.38	0.38	0.16	0.00	0.00	0.00	35.76	35.75	36.42	0.15	0.15	0.15
EWW	Mexico	MXN Carry	0.68	0.68	0.27	0.00	0.00	0.00	74.68	74.14	74.50	0.45	0.45	0.45
		MXN Future	0.64	0.64	0.24	0.00	0.00	0.00	72.43	72.35	71.59	0.41	0.41	0.41
EWY	Korea	KRW Carry	0.37	0.37	0.11	0.00	0.00	0.00	31.97	32.08	31.63	0.14	0.14	0.13
		KRW Future	0.59	0.59	0.19	0.00	0.00	0.00	49.86	50.63	43.33	0.34	0.35	0.28
EWZ	Brazil	BRL Carry	0.73	0.72	0.31	0.00	0.00	0.00	81.67	81.99	79.50	0.51	0.51	0.49
		BRL Future	0.68	0.68	0.28	0.00	0.00	0.00	73.09	73.62	71.61	0.45	0.46	0.44
EZA	South Africa	ZAR Carry	0.73	0.74	0.37	0.00	0.00	0.00	81.69	83.73	80.69	0.53	0.55	0.53
EZU	Euro	EUR Carry	0.38	0.39	0.15	0.00	0.00	0.00	32.86	33.67	35.85	0.14	0.15	0.17
		EUR Future	0.36	0.38	0.14	0.00	0.00	0.00	31.86	32.62	33.84	0.14	0.14	0.15
GLCR	Iceland	ISK Carry	0.41	0.43	0.16	0.00	0.00	0.00	6.71	6.68	5.79	0.15	0.15	0.12
INDA	India	INR Carry	0.38	0.39	0.11	0.00	0.00	0.00	24.97	25.28	26.11	0.15	0.15	0.16
		INR Future	0.57	0.57	0.17	0.00	0.00	0.00	39.57	40.09	40.99	0.32	0.33	0.34
MCHI	China	CNY Carry	0.22	0.22	0.03	0.00	0.00	0.00	14.11	14.17	17.03	0.05	0.05	0.07
		CNY Future	-0.21	-0.21	-0.04	0.00	0.00	0.00	-12.41	-12.39	-13.33	0.05	0.05	0.05
THD	Thailand	THB Carry	0.35	0.35	0.06	0.00	0.00	0.00	24.69	24.94	20.41	0.12	0.12	0.08
TUR	Turkey	TRY Carry	0.65	0.64	0.28	0.00	0.00	0.00	57.85	58.23	59.32	0.43	0.43	0.44

Table 7: Regression Results: β of instantaneously regression of sovereign equity returns against FX returns controlling for volatility target. β are scaled by -1 for carry indices since they represent long USD

ETF	Country	Vol. Target FX	Coef			p -value			t -value			R^2		
			Lagged	Perfect	None	Lagged	Perfect	None	Lagged	Perfect	None	Lagged	Perfect	None
ARGT	Argentina	ARS Carry	0.08	0.08	0.13	0.00	0.00	0.00	4.70	4.72	5.32	0.01	0.01	0.01
COLO	Colombia	COP Carry	0.32	0.32	0.27	0.00	0.00	0.00	21.79	22.08	33.86	0.10	0.10	0.21
ECH	Chile	CLP Carry	0.36	0.37	0.26	0.00	0.00	0.00	24.70	24.66	35.41	0.12	0.12	0.21
EDEN	Denmark	DKK Carry	0.33	0.33	0.17	0.00	0.00	0.00	18.07	18.09	20.91	0.08	0.08	0.11
EIDO	Indonesia	IDR Carry	0.20	0.21	0.09	0.00	0.00	0.00	12.29	12.73	20.49	0.04	0.04	0.10
EIS	Israel	ISR Future	0.37	0.37	0.23	0.00	0.00	0.00	19.69	19.89	27.74	0.08	0.08	0.15
ENOR	Norway	NOK Carry	0.47	0.47	0.37	0.00	0.00	0.00	26.47	26.53	41.20	0.16	0.16	0.32
		NOK Future	0.49	0.48	0.38	0.00	0.00	0.00	27.61	27.59	42.16	0.18	0.18	0.34
ENZL	New Zealand	NZD Carry	0.37	0.38	0.28	0.00	0.00	0.00	23.63	24.06	28.48	0.12	0.13	0.17
		NZD Future	0.40	0.40	0.30	0.00	0.00	0.00	25.72	26.11	29.61	0.15	0.15	0.19
EPHE	Philippines	PHP Carry	0.18	0.18	0.05	0.00	0.00	0.00	11.01	10.93	13.00	0.03	0.03	0.04
EPOL	Poland	PLN Carry	0.47	0.48	0.36	0.00	0.00	0.00	29.29	29.37	53.71	0.18	0.18	0.42
		PLN Future	0.47	0.48	0.36	0.00	0.00	0.00	29.67	29.73	53.96	0.18	0.18	0.43
EPU	Peru	PEN Carry	0.17	0.16	0.07	0.00	0.00	0.00	10.50	10.09	17.68	0.03	0.02	0.07
EWA	Australia	AUD Carry	0.40	0.40	0.31	0.00	0.00	0.00	32.29	32.62	49.63	0.12	0.12	0.25
		AUD Future	0.42	0.42	0.30	0.00	0.00	0.00	34.02	34.34	48.17	0.13	0.14	0.24
EWC	Canada	CAD Carry	0.33	0.34	0.17	0.00	0.00	0.00	24.90	25.07	31.53	0.08	0.08	0.12
		CAD Future	0.38	0.39	0.19	0.00	0.00	0.00	29.42	29.82	36.35	0.10	0.11	0.15
EWD	Sweden	SEK Carry	0.45	0.45	0.27	0.00	0.00	0.00	35.07	35.41	50.17	0.14	0.14	0.25
		SEK Future	0.58	0.58	0.35	0.00	0.00	0.00	36.60	36.71	52.41	0.18	0.18	0.32
EWH	Hong Kong	HKD Carry	0.05	0.05	0.00	0.00	0.00	0.00	3.78	3.68	2.94	0.00	0.00	0.00
EWJ	Japan	JPY Carry	0.31	0.32	0.18	0.00	0.00	0.00	24.19	24.51	28.37	0.07	0.07	0.10
		JPY Future	0.30	0.31	0.18	0.00	0.00	0.00	23.45	23.79	28.71	0.07	0.07	0.10
EWL	Switzerland	CHF Carry	0.49	0.49	0.26	0.00	0.00	0.00	41.46	41.74	39.20	0.19	0.19	0.17
		CHF Future	0.47	0.48	0.25	0.00	0.00	0.00	39.80	40.22	37.98	0.17	0.18	0.16
EWM	Malaysia	MYR Carry	0.20	0.19	0.04	0.00	0.00	0.00	9.51	9.14	14.14	0.01	0.01	0.03
EWS	Singapore	SGD Carry	0.30	0.30	0.07	0.00	0.00	0.00	21.11	21.04	22.95	0.06	0.06	0.07
EWT	Taiwan	TWD Carry	0.29	0.28	0.05	0.00	0.00	0.00	19.08	19.35	19.96	0.05	0.05	0.06
EWU	UK	GBP Carry	0.45	0.45	0.24	0.00	0.00	0.00	32.84	32.75	38.05	0.13	0.12	0.16
		GBP Future	0.45	0.45	0.23	0.00	0.00	0.00	32.83	32.80	36.90	0.13	0.13	0.15
EWW	Mexico	MXN Carry	0.31	0.31	0.26	0.00	0.00	0.00	22.18	21.92	44.14	0.07	0.07	0.22
		MXN Future	0.31	0.31	0.25	0.00	0.00	0.00	24.80	24.32	48.07	0.08	0.07	0.24
EWY	Korea	KRW Carry	0.39	0.39	0.17	0.00	0.00	0.00	27.38	27.77	35.57	0.10	0.11	0.16
		KRW Future	0.49	0.50	0.28	0.00	0.00	0.00	28.79	29.83	42.55	0.15	0.16	0.27
EWZ	Brazil	BRL Carry	0.42	0.43	0.37	0.00	0.00	0.00	33.82	34.16	69.73	0.15	0.15	0.43
		BRL Future	0.42	0.43	0.34	0.00	0.00	0.00	33.97	34.35	65.58	0.15	0.16	0.40
EZA	South Africa	ZAR Carry	0.45	0.47	0.46	0.00	0.00	0.00	32.60	33.93	67.67	0.15	0.17	0.44
EZU	Euro	EUR Carry	0.66	0.68	0.35	0.00	0.00	0.00	39.66	40.25	51.05	0.20	0.20	0.29
		EUR Future	0.65	0.68	0.34	0.00	0.00	0.00	39.78	40.46	49.71	0.20	0.20	0.28
GLCR	Iceland	ISK Carry	0.33	0.34	0.16	0.00	0.00	0.00	5.13	5.07	5.29	0.10	0.09	0.10
INDA	India	INR Carry	0.31	0.32	0.13	0.00	0.00	0.00	18.36	18.62	25.46	0.09	0.09	0.15
		INR Future	0.33	0.34	0.16	0.00	0.00	0.00	18.43	18.79	30.03	0.09	0.10	0.22
MCHI	China	CNY Carry	0.17	0.17	0.04	0.00	0.00	0.00	10.08	9.97	14.22	0.03	0.03	0.05
		CNY Future	-0.17	-0.17	-0.04	0.00	0.00	0.00	-9.41	-9.18	-12.20	0.03	0.03	0.04
THD	Thailand	THB Carry	0.28	0.28	0.08	0.00	0.00	0.00	17.63	17.73	20.48	0.06	0.06	0.08
TUR	Turkey	TRY Carry	0.33	0.32	0.30	0.00	0.00	0.00	24.68	24.43	49.86	0.12	0.12	0.35

Table 8: Regression Results: β of instantaneously regression of sovereign equity returns differential against FX returns controlling for volatility target. β are scaled by -1 for carry indices since they represent long USD

Now compare both returns and returns differential for FX returns. The idea is that by running a regression of ETF returns and ETF returns differential will give an idea of how much each factor contributes while controlling for eachother. See the next page for the plot of the β and t -values of the β which show ETF returns tend to have higher β s relative to returns differential implying that they are large contributor.

When comparing the two regressions in terms of β and t -value of β when they are regressed together show that there is a stronger preference toward the raw returns as opposed to the return's differential. Below is a scatter of the values comparing the returns and returns differential exogenous variables parameters. There is a diagonal line which shows the tradeoff between parameters. Parameters below the line have a stronger relationship to the ETF returns and above the line infers the opposite. Volatility hedging is taken into consideration as opposed to raw returns.

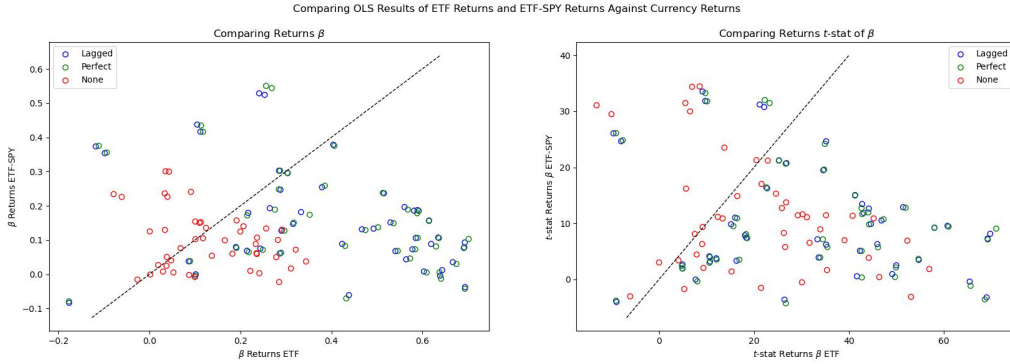


Figure 2: Comparing Parameters of $R_{fx} \sim R_{ETF} + R_{ETF-SPY}$

The β are quite high for some of the regressions. After some testing there isn't a clear explanation for why some regressions have high β . The results are re-tested with adding lagged differences of FX returns and changes in the first 3 Principal Components of each country's respective yield curve. Some of the yield curve data for certain countries is missing and therefore all of the countries can't be testing. Even after controlling for the principal components and lagged fx differences the β are quite high

Param	Group	Baseline	PCs & Lagged
ETF Return	EM Carry	0.30	0.37
	G10 Carry	0.32	0.30
	Illiquid Futures	0.31	0.35
	Liquid Futures	0.32	0.30
ETF Spread Return	EM Carry	0.11	0.15
	G10 Carry	0.20	0.24
	Illiquid Futures	0.15	0.16
	Liquid Futures	0.19	0.24

Table 9: Regression Results: β of instantaneously regression of FX returns against ETF and ETF-SPY returns, controlling for lagged FX returns and changes in first 3 Principal Components of Yield Curve (if applicable)

5 Optimized Signal Backtest

From the factor exposure section a couple of signals are implemented. They are ran individually as opposed to a multi-factor model. The two factors of interest to trade are

1. The sovereign ETF returns spread trend (z-score)
2. The sovereign ETF returns differential trend (z-score)

Both of these factors have been researched extensively throughout this paper. They have been constructed using both raw returns and volatility targeted returns, which makes a considerable difference in outcome. Therefore there are four signals to trade signal name (Returns, and Returns Spread) and signal type (Raw Returns and Volatility Targeted returns). Rather than trading the the full spectrum of each factor they are optimized, this optimization does the following

1. Only trading signals when they are screening at the largest magnitude
2. Reducing correlation
3. Harvesting non-linear alpha
4. Parametric fit

The optimization follows, per each signal decile the values historically into bins. Then calculate the sharpe per each decile bin. Then examine the outer sets of bins 1st & 2nd (lower group) and 9th & 10th (upper group). Then examine whether or not these lower/upper groups have the same sharpe, and if so take a position of the same sign. This is done full-sample in-sample to begin with. The optimization inherently works in full sample studies and finds alpha (spuriously) with a large enough sample size. Therefore the out-of-sample and in-sample to out-of-sample comparison is much more important.

Group	Signal Name	Vol. Target Sample Signal	Lagged		Contemporaneous	
			In-Sample	Out-Sample	In-Sample	Out-Sample
EM Carry	ETF Returns	Raw Return	3.07	2.58	3.26	2.91
		Vol. Adj	2.62	2.53	2.85	2.79
	ETF Returns Differential	Raw Return	1.98	1.69	2.12	1.88
		Vol. Adj	1.58	1.39	1.66	1.47
Full Portfolio	ETF Returns	Raw Return	2.68	2.43	2.78	2.65
		Vol. Adj	2.49	2.46	2.64	2.59
	ETF Returns Differential	Raw Return	2.12	1.67	2.21	1.81
		Vol. Adj	1.68	1.24	1.75	1.35
G10 Carry	ETF Returns	Raw Return	0.86	0.58	0.86	0.56
		Vol. Adj	0.88	0.34	0.86	0.37
	ETF Returns Differential	Raw Return	0.78	0.46	0.80	0.48
		Vol. Adj	0.72	0.37	0.73	0.38
Illiquid Fut	ETF Returns	Raw Return	2.21	2.04	2.23	2.14
		Vol. Adj	2.20	2.25	2.25	2.36
	ETF Returns Differential	Raw Return	1.16	1.26	1.16	1.34
		Vol. Adj	1.12	0.67	1.15	0.55
Liquid Fut	ETF Returns	Raw Return	1.48	1.20	1.52	1.24
		Vol. Adj	1.31	1.19	1.32	1.24
	ETF Returns Differential	Raw Return	1.02	0.63	1.01	0.77
		Vol. Adj	0.86	0.51	0.90	0.66

Table 10: Backtest Results: Sharpe of backtested portfolios using decile optimization in-sample (full sample) and expanding out-of-sample (updated weekly)

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